

MACHINEPULSE · DESIGN DECISION RECORD

MachinePulse TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	MPL-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/machinepulse		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for D1 to D10. On 2026-09-25 Amish wrote "i accept all your recommendations, go with them across all repos", so items D1 to D10 are decided as recommended (see MPL-DDR-002). Items O1 and O2 carry no single recommendation and remain "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis MPL-PRC-001 v0.2 listed key design choices, most with options and a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction, open for his review. Items without a single recommendation stay open.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in MPL-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items decided by Amish on 2026-09-25 (first adopted for TRL 3 work, then accepted).

#	Item	Adopted recommendation	Status
D1	Radio	Wi-Fi on the ESP32-S3 first; a summaries-only LoRaWAN variant through the FieldNode radio core stays a later option	Decided by Amish, 2026-09-25: go with recommendation
D2	Power	Certified 5 V USB adapter; no battery and no energy harvesting	Decided by Amish, 2026-09-25: go with recommendation
D3	Accelerometer	IIS3DWB class, with the ADXL345 class kept as a documented fallback	Decided by Amish, 2026-09-25: go with recommendation
D4	Current sensing	One CT on one phase, for single-phase and three-phase machines	Decided by Amish, 2026-09-25: go with recommendation
D5	Alert logic	Baseline learned over the first week per load band; flags at twice baseline vibration or a temperature rise above baseline, sent to a person; fixed ISO 20816 zones a later option	Decided by Amish, 2026-09-25: go with recommendation
D6	Data home	Twinkl gateway by default, any MQTT broker supported, no cloud dependency	Decided by Amish, 2026-09-25: go with recommendation
D7	Enclosure	Stock IP54 ABS box for the first build, with a stand-off or insulating pad studied at TRL 3 for hot frames	Decided by Amish, 2026-09-25: go with recommendation
D8	Pilot site	A makerspace or university workshop with a lathe and a compressor	Decided by Amish, 2026-09-25: go with recommendation
D9	Budget	No change: <code>budget_usd</code> stays \$80. Superseded in effect by MPL-DDR-002, N1, which raises it to \$81 for high-temperature magnets	Decided by Amish, 2026-09-25: go with recommendation
D10	Data schedule	Features computed on the pod every minute, a spectrum every 15 min, raw bursts on demand	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remain open.

#	Item	Status
01	Co-design partner for the alerts and dashboard: who decides what "changed" should mean to an operator. No recommendation was made.	Proposed, awaiting Amish
02	How to treat R4 with one CT. The TRL 2 note offered two courses without choosing: accept that R4 is not met in the first build, or relax R4 to "relative energy trend". MPL-CAL-001 now puts R4 at risk (8.5 % RSS) rather than not met, which bears on this choice. R4 keeps its 10 % target until Amish decides.	Proposed, awaiting Amish

■ Consequences

- `project.yaml` : only the TRL fields change. No budget change was recommended (D9), and no reworded pitch or problem line was recommended, so `budget_usd`, `pitch` and `problem` and the matching lines in `README.md` keep their wording. MPL-DDR-002 later raised `budget_usd` to \$81.
- MPL-PRB-001, MPL-PRC-001 and MPL-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed". No requirement target changes as a result of these decisions; R4 stays at 10 % pending O2.
- D7 study result (MPL-CAL-001, section G): 5 mm nylon stand-offs between the sensor block and the box floor keep the floor at 53.9 °C on an 80 °C frame, against 67.2 °C without them. They are added to the design and the BOM (line 13, \$1.00), which brings the parts to \$80.00, exactly the budget.
- Other details added within these decisions by the calculations: a CT bias of 1.50 V, clamp diodes on the CT input, a ± 16 g accelerometer range, a module without octal PSRAM, 3 mm steel pads, a silicone boot round the boss and a load-dependent power factor curve in the gateway software.
- New proposals from MPL-CAL-001 (high-temperature magnets, R8 range, R3 and R5 follow-up) were decided by Amish on 2026-09-25 and are recorded in MPL-DDR-002.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.